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POWER TOOL WITH IMPULSE ASSEMBLY

Abstract

A power tool including a motor, an impulse assembly configured to be driven by the motor, a temperature sensor, and a controller. The temperature sensor is configured to output a signal related to a temperature of the power tool. The temperature of the power tool is correlated to a temperature of a fluid within the impulse assembly. The controller is connected to the temperature sensor. The controller is configured to determine the temperature of the power tool based the signal from the temperature sensor, and control, in response to the temperature of the power tool being greater than a temperature threshold, the power tool based on the temperature of the power tool.

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Background/Summary

RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/560,273, filed Mar. 1, 2024, and U.S. Provisional Patent Application No. 63/658,143, filed Jun. 10, 2024, the entire content of each of which is hereby incorporated by reference.

FIELD

[0002] Embodiments described herein relate to power tools.

SUMMARY

[0003] Impulse power tools are capable of delivering rotational impacts to a workpiece at high speeds by storing energy in a rotating mass and transmitting it to an output shaft. Such impulse power tools generally have an output shaft, which may or may not be capable of holding a tool bit or engaging a socket. Impulse tools generally utilize the percussive transfers of high momentum, which is transmitted through the output shaft using a variety of technologies, such as electric, oil-pulse, mechanical-pulse, or any suitable combination thereof.

[0004] Power tools described herein include a housing, a motor positioned within the housing, an impulse assembly coupled to the motor to receive torque therefrom, the impulse assembly including a cylinder at least partially forming a chamber containing a hydraulic fluid, an anvil positioned at least partially within the chamber, and a hammer positioned at least partially within the chamber and engageable with the anvil for transferring rotational impacts to the anvil, the hammer including a through hole, and a valve configured to control flow of the hydraulic fluid through the through hole.

[0005] When the hydraulic fluid reaches a certain temperature, an impulse tool may experience thermal failure. In some instances, it is desirable to shut down the impulse tool before the tool experiences thermal failure. However, it is difficult to measure the temperature of the hydraulic fluid directly. Therefore, embodiments described herein provide a system and process for controlling an impulse tool using a determined temperature that correlates to a temperature of a fluid within the impulse mechanism. The determined temperature may be based on one or more signals from temperature sensors away from a path of airflow circulation in the impulse tool (for example, one or more temperature sensors positioned on a light assembly PCB) or one or more temperature sensors positioned on a Hall effect sensor PCB.

[0006] Power tools described herein include a motor, an impulse assembly configured to be driven by the motor, a temperature sensor, and a controller. The temperature sensor is configured to output a signal related to a temperature of the power tool. The temperature of the power tool is correlated to a temperature of a fluid within the impulse assembly. The controller is connected to the temperature sensor. The controller is configured to determine the temperature of the power tool based the signal from the temperature sensor, and control, in response to the temperature of the power tool being greater than a temperature threshold, the power tool based on the temperature of the power tool.

[0007] In some aspects, to control the power tool based on the temperature of the power tool, the controller is configured to one of shut down the power tool or reduce an output of the power tool.

[0008] In some aspects, the temperature sensor is positioned away from a path of airflow circulation within the power tool.

[0009] In some aspects, the temperature sensor is positioned on a light assembly printed circuit board (“PCB”).

[0010] In some aspects, the temperature sensor is positioned on a Hall effect sensor printed circuit board (“PCB”).

[0011] In some aspects, the controller is further configured to compare, in response to the temperature of the power tool being less than the temperature threshold and greater than or equal to a second temperature threshold, an amount of time for which the temperature of the power tool is less than first temperature threshold and greater than or equal to the second temperature threshold to a time threshold, and disable, in response to the amount of time reaching the time threshold, the power tool.

[0012] Power tools described herein include a motor, an impulse assembly configured to be driven by the motor, a first temperature sensor, a second temperature sensor, and a controller. The first temperature sensor is configured to output a first signal related to a temperature of the power tool. The temperature of the power tool is correlated to a temperature of a fluid within the impulse assembly. The second temperature sensor is configured to output a second signal related to the temperature of the power tool. The controller is connected to the first temperature sensor and the second temperature sensor. The controller is configured to determine the temperature of the power tool based the first signal from the first temperature sensor and the second signal from the second temperature sensor, and control, in response to the temperature of the power tool being greater than a temperature threshold, the power tool based on the temperature of the power tool.

[0013] In some aspects, to control the power tool based on the temperature of the power tool, the controller is configured to shut down the power tool.

[0014] In some aspects, the first temperature sensor and the second temperature sensor are positioned away from a path of airflow circulation within the power tool.

[0015] In some aspects, the first temperature sensor is positioned on a light assembly printed circuit board (“PCB”).

[0016] In some aspects, the second temperature sensor is positioned on a Hall effect sensor PCB.

[0017] In some aspects, the controller is further configured to compare, in response to the temperature of the power tool being less than the temperature threshold and greater than or equal to a second temperature threshold, an amount of time for which the temperature of the power tool is less than first temperature threshold and greater than or equal to the second temperature threshold to a time threshold, and disable, in response to the amount of time reaching the time threshold, the power tool.

[0018] Methods of controlling a power tool described herein include receiving, from a temperature sensor, a signal related to a temperature of the power tool, the temperature of the power tool being correlated to a temperature of a fluid within an impulse assembly, determining the temperature of the power tool based the signal from the temperature sensor, and controlling, in response to the temperature of the power tool being greater than a temperature threshold, the power tool based on the temperature of the power tool.

[0019] In some aspects, controlling the power tool based on the temperature of the power tool includes one or shutting down the power tool or reducing an output of the power tool.

[0020] In some aspects, the temperature sensor is positioned away from a path of airflow circulation within the power tool.

[0021] In some aspects, the temperature sensor is positioned on a light assembly printed circuit board (“PCB”).

[0022] In some aspects, the temperature sensor is positioned on a Hall effect sensor printed circuit board (“PCB”).

[0023] In some aspects, the method further includes comparing, in response to the temperature of the power tool being less than the temperature threshold and greater than or equal to a second temperature threshold, an amount of time for which the temperature of the power tool is less than first temperature threshold and greater than or equal to the second temperature threshold to a time threshold, and disabling, in response to the amount of time reaching the time threshold, the power tool.

[0024] In some aspects, the temperature sensor is positioned within the impulse assembly.
[0025] In some aspects, the method further includes receiving, from a second temperature sensor, a second signal related to the temperature of the power tool, the temperature of the power tool being correlated to the temperature of the fluid within an impulse assembly.
[0026] Before any embodiments are explained in detail, it is to be understood that the embodiments are not limited in application to the details of the configurations and arrangements of components set forth in the following description or illustrated in the accompanying drawings. The embodiments are capable of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of "including," "comprising," or "having" and variations thereof are meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms "mounted," "connected," "supported," and "coupled" and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings.
[0027] Unless the context of their usage unambiguously indicates otherwise, the articles "a," "an," and "the" should not be interpreted as meaning "one" or "only one." Rather these articles should be interpreted as meaning "at least one" or "one or more." Likewise, when the terms "the" or "said" are used to refer to a noun previously introduced by the indefinite article "a" or "an," "the" and "said" mean "at least one" or "one or more" unless the usage unambiguously indicates otherwise.
[0028] In addition, it should be understood that embodiments may include hardware, software, and electronic components or modules that, for purposes of discussion, may be illustrated and described as if the majority of the components were implemented solely in hardware. However, one of ordinary skill in the art, and based on a reading of this detailed description, would recognize that, in at least one embodiment, the electronic-based aspects may be implemented in software (e.g., stored on non-transitory computer-readable medium) executable by one or more processing units, such as a microprocessor and/or application specific integrated circuits ("ASICs"). As such, it should be noted that a plurality of hardware and software based devices, as well as a plurality of different structural components, may be utilized to implement the embodiments. For example, "servers," "computing devices," "controllers," "processors," etc., described in the specification can include one or more processing units, one or more computer-readable medium modules, one or more input/output interfaces, and various connections (e.g., a system bus) connecting the components.
[0029] Relative terminology, such as, for example, "about," "approximately," "substantially," etc., used in connection with a quantity or condition would be understood by those of ordinary skill to be inclusive of the stated value and has the meaning dictated by the context (e.g., the term includes at least the degree of error associated with the measurement accuracy, tolerances [e.g., manufacturing, assembly, use, etc.] associated with the particular value, etc.). Such terminology should also be considered as disclosing the range defined by the absolute values of the two endpoints. For example, the expression "from about 2 to about 4" also discloses the range "from 2 to 4". The relative terminology may refer to plus or minus a percentage (e.g., 1%, 5%, 10%) of an indicated value.
[0030] It should be understood that although certain drawings illustrate hardware and software located within particular devices, these depictions are for illustrative purposes only. Functionality described herein as being performed by one component may be performed by multiple components in a distributed manner. Likewise, functionality performed by multiple components may be consolidated and performed by a single component. In some embodiments, the illustrated components may be combined or divided into separate software, firmware and/or hardware. For example, instead of being located within and performed by a single electronic processor, logic and processing may be distributed among multiple electronic processors. Regardless of how they are combined or divided, hardware and software components may be located on the same computing device or may be distributed among different computing devices connected by one or more

networks or other suitable communication links. Similarly, a component described as performing particular functionality may also perform additional functionality not described herein. For example, a device or structure that is "configured" in a certain way is configured in at least that way but may also be configured in ways that are not explicitly listed.
[0031] Accordingly, in the claims, if an apparatus, method, or system is claimed, for example, as including a controller, control unit, electronic processor, computing device, logic element, module, memory module, communication channel or network, or other element configured in a certain manner, for example, to perform multiple functions, the claim or claim element should be interpreted as meaning one or more of such elements where any one of the one or more elements is configured as claimed, for example, to make any one or more of the recited multiple functions, such that the one or more elements, as a set, perform the multiple functions collectively.
[0032] Other aspects of the disclosure will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0033] FIG. **1** is a cross-sectional view of a power tool, according to some embodiments.
[0034] FIG. **2** is a cross-sectional view of an impulse assembly for the power tool of FIG. **1**, according to some embodiments.
[0035] FIG. **3** is a schematic illustrating operation of the impulse assembly of FIG. **2**, according to some embodiments.
[0036] FIG. **4** is a cross-sectional view of the power tool of FIG. **1**, according to some embodiments.
[0037] FIG. **5** illustrates a motor and a Hall effect sensor printed circuit board for the power tool of FIG. **1**, according to some embodiments.
[0038] FIG. **6** illustrates a light assembly printed circuit board for the power tool of FIG. **1**, according to some embodiments.
[0039] FIG. **7** is a cross-sectional view of the power tool of FIG. **1**, according to some embodiments.
[0040] FIG. **8** illustrates a control system for the power tool of FIG. **1**, according to some embodiments.
[0041] FIG. **9** illustrates a process for controlling the power tool of FIG. **1**, according to some embodiments.
[0042] FIG. **10** illustrates a process for controlling the power tool of FIG. **1**, according to some embodiments.
[0043] FIG. **11** is a graph illustrating operation of the power tool of FIG. **1**, according to some embodiments.
[0044] FIG. **12** illustrates a process for controlling the power tool of FIG. **1**, according to some embodiments.
[0045] FIG. **13** is a graph illustrating operation of the power tool of FIG. **1**, according to some embodiments.
DETAILED DESCRIPTION
[0046] With reference to FIG. **1**, a power tool (e.g., an impulse tool **10**) embodying aspects of the present disclosure is shown. The impulse tool **10** includes a main housing **14** and a rotational impulse assembly **18** (see FIG. **2**) positioned within the main housing **14**. The impulse tool **10** also includes an electric motor **22** (e.g., a brushless direct current motor) coupled to the impulse assembly **18** to provide torque thereto and positioned within the main housing **14**, and a transmission (e.g., a single or multi-stage planetary transmission) positioned between the motor **22**

and the impulse assembly **18**. In some embodiments, the impulse tool **10** is battery-powered and is configured to be powered by a rechargeable power tool battery pack.

[0047] With reference to FIGS. **1** and **2**, the illustrated impulse assembly **18** includes a hammer **26** and an anvil **30**. A driven end **26***a* of the hammer **26** is coupled to the electric motor **22** to receive torque therefrom, causing the hammer **26** to rotate. In some embodiments, a transmission, such as a planetary transmission, may be provided between the electric motor **22** and the hammer **26** to provide a speed reduction and torque increase from the electric motor **22** to the hammer **26**. In such embodiments, the hammer **26** may be coupled to or may integrally include a carrier of the planetary transmission.

[0048] Referring to FIG. **2**, the hammer **26** includes hammer lugs **34** that are configured to impact the anvil **30** as the motor **22** drives rotation of the hammer **26**. Specifically, the illustrated hammer **26** includes two hammer lugs **34**. The hammer lugs **34** extend inward from an inner circumferential surface **38** of the hammer **26**. The hammer **26** at least partially defines a hammer chamber **42** that contains an incompressible fluid (e.g., hydraulic fluid, oil, etc.). The hammer chamber **42** is sealed and is also partially defined by an end cap **46** threadedly secured to a front end **26***b* of the hammer **26** that is opposite from the driven end **26***a* (FIG. **1**). The hydraulic fluid in the hammer chamber **42** may reduce the wear and the noise of the impulse assembly **18** that is created by impacting the hammer **26** and the anvil **30**.

[0049] The anvil **30** is positioned at least partially within the hammer chamber **42** and includes a blade assembly **50** and an output shaft **54** with a hexagonal receptacle **58** therein for receipt of a tool bit. The blade assembly **50** includes blades **62***a*, **62***b* that are configured to receive impacts from the hammer lugs **34** and a spring **66** that biases each of the blades **62***a*, **62***b* into engagement with the inner circumferential surface **38** of the hammer **26**.

[0050] The illustrated blade assembly **50** includes two blades **62***a*, **62***b* such that each blade **62***a*, **62***b* is configured to receive an impact from a corresponding one of the hammer lugs **34**. Each of the blades **62***a*, **62***b* extends through the output shaft **54** such that the blades **62***a*, **62***b* are configured to transfer rotational impacts from the hammer lugs **34** to the output shaft **54**. In the illustrated embodiment, the blade assembly **50** includes a single spring **66** that biases both of the blades **62***a*, **62***b*. In some embodiments, the blade assembly **50** may include two springs such that each spring biases a corresponding blade **62***a*, **62***b*. The output shaft **54** extends from the hammer chamber **42** and through the end cap **46**. The output shaft **54** extends along and is configured to rotate about an axis Al. The output shaft **54** defines an anvil chamber **70** and fill holes **74** that place the anvil chamber **70** in fluid communication with the hammer chamber **42**. The fill holes **74** may be intermittently opened and closed to enable and inhibit flow of the hydraulic fluid between the hammer chamber **42** and the anvil chamber **70**.

[0051] With reference to FIGS. **1-3**, at the start of operation of the impulse tool **10**, the fill holes **74** in the output shaft **54** are open and allow hydraulic fluid to flow freely between the hammer chamber **42** and the anvil chamber **70**. As such, the pressure in the hammer chamber **42** and the pressure in the anvil chamber **70** are equal at the start of operation. During operation, each of the hammer lugs **34** intermittently applies rotational impacts on a corresponding one of the blades **62***a*, **62***b* to impart consecutive rotational impacts to the output shaft **54**. As the hammer lugs **34** begin to strike the blades **62***a*, **62***b* (i.e., a pulse start), the fill holes **74** may close and pressure of the hydraulic fluid within the anvil chamber **70** may rapidly increase such that the pressure in anvil chamber **70** is relatively higher than the pressure of hydraulic fluid in the hammer chamber **42**. As impact progresses, the hammer lugs **34** may ramp up and slip over the blades **62***a*, **62***b*, thereby pushing the blades **62***a*, **62***b* against the bias of the spring **66** and against the relatively higher pressure of fluid in the anvil chamber **70**. The fluid within the anvil chamber **70** may be forced out into the hammer chamber **42** through a restricted orifice to allow the blades **62***a*, **62***b* to move inwardly towards the axis Al. As the blades **62***a*, **62***b* move inwardly, the relatively higher pressure within the anvil chamber **70** may begin to be relieved through the orifice such that fluid in the hammer chamber **42** and the anvil chambers **70** approach equal levels. The hydraulic fluid may thus slow and damp inward movement of the blades **62***a*, **62***b* and thereby increase the duration over which the hammer lugs **34** engage the blades **62***a*, **62***b*. The hydraulic fluid may also reduce noise emissions and inhibit wear on the components of the hammer **26** and the anvil **30** throughout impact.

[0052] Once the hammer lugs **34** pass over the blades **62***a*, **62***b* (e.g., a pulse end), the springs **66** bias the blades **62***a*, **62***b* outwardly from the axis A**1** and back into engagement with the inner circumferential surface **38** of the hammer **26**. Additionally, the fill holes **74** may open again, allowing hydraulic fluid to pass between the hammer chamber **42** and the anvil chamber **70**. As the spring **66** biases the blades **62***a*, **62***b* outwardly, the pressure within the anvil chamber **70** may be relatively lower than the pressure in the hammer chamber **42**. As such, the hydraulic fluid may flow from the hammer chamber **42** to the anvil chamber **70** until the pressures in each chamber **42**, **70** are equal. With the blades **62***a*, **62***b* biased into engagement with the inner circumferential surface **38** of the hammer **26** and the pressures within each chamber **42**, **70** equal, the impulse assembly **18** is ready for another impact and for the cycle to repeat.

[0053] FIG. **4** illustrates a cross-sectional view of the impulse tool **10**. As shown in FIG. **4**, the impulse tool **10** includes a first temperature sensor (e.g., a thermistor, a thermocouple, etc.) or a first plurality of temperature sensors (e.g., a thermistor, a thermocouple, etc.) **400** mounted to a Hall effect sensor printed circuit board (“PCB”). The Hall effect sensor PCB can be mounted to a stator of a motor within the impulse tool **10**. The impulse tool **10** also includes a second temperature sensor (e.g., a thermistor) or a second plurality of temperature sensors (e.g., thermistors) **405** mounted to a light assembly printed circuit board (“PCB”). In some embodiments, the temperature sensor **405** is mounted to a bolt plate. The temperature sensors **400**, **405** can be used to indirectly measure a temperature of a fluid in the impulse assembly **18** or, in other words, determine a temperature that is correlated with the temperature of the fluid in the impulse assembly **18**. FIG. **4** also illustrates airflow paths **410**, **415** for the impulse tool **10**, which can affect the temperature measurements using the temperature sensor **405**. In some embodiments, a gearcase **420** for the transmission includes an additional temperature sensor **425** (e.g., a thermistor, a thermocouple, etc.) coupled to the gearcase **420**.

[0054] FIG. **5** illustrates a motor **500** adjacent to a Hall effect sensor PCB **505** for the impulse tool **10**. The Hall effect sensor PCB **505** includes a plurality of Hall effect sensors **510** positioned, for example, on a first side of the Hall effect sensor PCB **505**. The Hall effect sensor PCB **505** includes one or more temperature sensors (e.g., thermistors, thermocouples, etc.) **515**. In some embodiments, a single temperature sensor is implemented. In other embodiments, a plurality of temperature sensors are implemented. The temperature sensors **515** can be used to measure a temperature of a fluid in the impulse assembly **18**. It should be noted that while each of the temperature sensors **515** are illustrated in FIG. **5** as being positioned on the first side of the PCB, in some embodiments, one or more of the temperature sensors **515** may be positioned on a second, opposite side of the PCB. In some embodiments, the temperature sensors **515** may be positioned on only the second side of the Hall effect sensor PCB.

[0055] FIG. **6** illustrates a nose cone **600** for the impulse tool **10** that surrounds a light assembly PCB **605**. The light assembly PCB **605** includes a plurality of lights or LEDs **610** positioned, for example, on a first side of the light assembly PCB **605**. The light assembly PCB **605** includes one or more temperature sensors (e.g., thermistors, thermocouples, etc.) **615**. In some embodiments, a single temperature sensor is positioned on the light assembly PCB **605**. In other embodiments, a plurality of temperature sensors are positioned on the light assembly PCB **605**. The temperature sensors **615** can be used to measure a temperature of a fluid in the impulse assembly **18**. In some embodiments, the temperature sensors **615** are removed from or positioned away from a path of airflow circulation in the impulse tool **10**.

[0056] FIG. **7** illustrates the impulse assembly **18** of the impulse tool **10**. In the embodiment illustrated in FIG. **7**, one or more temperature sensors (e.g., thermistors) **700** are positioned within the hammer chamber **42** of the impulse assembly **18**.

[0057] As shown in FIG. **8**, the impulse tool **10** includes the motor **22**. The motor **22** actuates a drive device and allows the drive device to perform the particular task (e.g., provide a rotational output to drive a fastener). The motor **22** may include a rotor and a stator. The rotor may be coupled to a motor shaft to produce a rotational output to the output drive device directly or via one or more gears. A primary power source (e.g., a battery pack) **810** couples to the impulse tool **10** and provides electrical power to energize the motor **22**. The motor **22** is energized based on the position of a trigger **830**. When the trigger **830** is depressed the motor **22** is energized, and when the trigger **830** is released, the motor **22** is de-energized. In the illustrated embodiment, the trigger **830** extends partially down a length of a handle. In other embodiments, the trigger **830** extends down the entire length of the handle or may be positioned elsewhere on the impulse tool **10**. The trigger **830** is moveably coupled to the handle such that the trigger **830** moves with respect to the tool housing. The trigger **830** is coupled to a push rod, which is engageable with a trigger switch **835**. The trigger **830** moves in a first direction towards the handle when the trigger **830** is depressed by the user. The trigger **830** is biased (e.g., with a spring) such that it moves in a second direction away from the handle, when the trigger **830** is released by the user. When the trigger **830** is depressed by the user, the push rod activates the trigger switch **835**, and when the trigger **830** is released by the user, the trigger switch **835** is deactivated. In some embodiments, the trigger **1130** is disconnected from the electrical trigger switch **835**. In such embodiments, the trigger switch **835** may include, for example, a transistor. Additionally, for such electronic embodiments, the trigger **830** may not include a push rod to activate the mechanical switch. Rather, the electrical trigger switch **835** may be activated by, for example, a position sensor (e.g., a Hall effect sensor) that relays information about the relative position of the trigger **830** to the tool housing or electrical trigger switch **835**. The trigger switch **835** outputs a signal indicative of the position of the trigger **830**. In some instances, the signal is binary and indicates either that the trigger **830** is depressed or released. In other instances, the signal indicates the position of the trigger **830** with more precision. For example, the trigger switch **835** may output an analog signal that varies from 0 to 5 volts depending on the extent that the trigger **830** is depressed. For example, 0 V output indicates that the trigger **830** is released, 1 V output indicates that the trigger **830** is 20% depressed, 2 V output indicates that the trigger **830** is 40% depressed, 3 V output indicates that the trigger **830** is 60% depressed, 4 V output indicates that the trigger **830** is 80% depressed, and 5 V indicates that the trigger **830** is 100% depressed. The signal output by the trigger switch **835** may be analog or digital.

[0058] As also shown in FIG. **8**, the impulse tool **10** includes a switching network **815**, sensors **820**, indicators **840**, a battery pack interface **805**, a power input unit **860**, an electronic controller **800**, a wireless communication controller **850**, and a back-up power source **855**. The back-up power source **855** includes, in some embodiments, a coin cell battery or another similar small replaceable power source. The battery pack interface **805** is coupled to the electronic controller **800** and couples to the battery pack **810**. The battery pack interface **805** includes a combination of mechanical (e.g., a battery pack receiving portion) and electrical components configured to and operable for interfacing (e.g., mechanically, electrically, and communicatively connecting) the impulse tool **10** with the battery pack **810**. The battery pack interface **805** is coupled to the power input unit **860**. The battery pack interface **805** transmits the power received from the battery pack **810** to the power input unit **860**. The power input unit **1160** includes active and/or passive components (e.g., voltage step-down controllers, voltage converters, rectifiers, filters, etc.) to regulate or control the power received through the battery pack interface **805** and to the wireless communication controller **850** and controller **800**.

[0059] The switching network **815** enables the electronic controller **800** to control the operation of the motor **22**. Generally, when the trigger **830** is depressed as indicated by an output of the trigger switch **835**, electrical current is supplied from the battery pack interface **805** to the motor **22**, via the switching network **815**. When the trigger **830** is not depressed, electrical current is not supplied from the battery pack interface **805** to the motor **22**.

[0060] In response to the electronic controller **800** receiving the activation signal from the trigger switch **835**, the electronic controller **800** activates the switching network **815** to provide power to the motor **22**. The switching network **815** controls the amount of current available to the motor **22** and thereby controls the speed and torque output of the motor **22**. The switching network **815** may include numerous FETs, bipolar transistors, or other types of electrical switches. For instance, the switching network **1115** may include a six-FET bridge that receives pulse-width modulated ("PWM") signals from the electronic controller **800** to drive the motor **22**.

[0061] The sensors **820** are coupled to the electronic controller **800** and communicate to the electronic controller **800** various signals indicative of different parameters of the impulse tool **10** or the motor **22**. The sensors **820** include Hall effect sensors, among other sensors, such as, for example, one or more voltage sensors, one or more temperature sensors (e.g., temperature sensors **400**, **405**, **515**, **615**, **700**), one or more torque sensors, etc. One or more signals from the temperature sensors **400**, **405**, **515**, **615**, **700** can be used by the controller **800** to determine a temperature correlated to a temperature of the fluid within the impact mechanism. For example, the temperature of the fluid can be determined or estimated is based on testing data used to correlate those temperatures determined to internal fluid data (e.g., bench testing, manual measurement of fluid temperature, etc.). In some embodiments, the temperature of the fluid is not directly measured (i.e., indirect measurement). In other embodiments, the temperature of the fluid can be directly measured (e.g., using temperature sensors **700**). Each Hall effect sensor outputs motor feedback information to the electronic controller **800**, such as an indication (e.g., a pulse) when a magnet of the motor's rotor rotates across the face of that Hall effect sensor. Based on the motor feedback information from the Hall effect sensors, the electronic controller **800** can determine the position, velocity, and acceleration of the rotor. In response to the motor feedback information and the signals from the trigger switch **835**, the electronic controller **800** transmits control signals to control the switching network **815** to drive the motor **22**. For instance, by selectively enabling and disabling the FETs of the switching network **815**, power received via the battery pack interface **805** is selectively applied to stator coils of the motor **22** to cause rotation of its rotor. The motor feedback information is used by the electronic controller **800** to ensure proper timing of control signals to the switching network **815** and, in some instances, to provide closed-loop feedback to control the speed of the motor **22** to be at a desired level.

[0062] The indicators **840** are also coupled to the electronic controller **800** and receive control signals from the electronic controller **800** to turn on and off or otherwise convey information based on different states of the impulse tool **10**. The indicators **840** include, for example, one or more light-emitting diodes ("LED") or a display screen. The indicators **840** can be configured to display conditions of, or information associated with, the impulse tool **10**. For example, the indicators **840** are configured to indicate measured electrical characteristics of the impulse tool **10**, the status of the impulse tool **10**, the mode of the power tool, etc. The indicators **840** may also include elements to convey information to a user through audible or tactile outputs. Work light LEDs **845** are also controllable by the electronic controller **800**, for example, to illuminate in response to the trigger **830** being actuated.

[0063] As described above, the electronic controller **800** is electrically and/or communicatively connected to a variety of components of the impulse tool **10**. In some embodiments, the electronic controller **800** includes a plurality of electrical and electronic components that provide power, operational control, and protection to the components within the electronic controller **800** and/or impulse tool **10**. For example, the electronic controller **800** includes, among other things, a processing unit **865** (e.g., a microprocessor, a microcontroller, an electronic controller, an electronic processor, or another suitable programmable device), a memory **870**, input units **875**, and output units **880**. The processing unit **865** (herein, electronic processor **865**) includes, among other things, a control unit **885**, an arithmetic logic unit ("ALU") **890**, and a plurality of registers **895** (shown as a group of registers in FIG. **8**). In some embodiments, the electronic controller **800** is implemented partially or entirely on a semiconductor (e.g., a field-programmable gate array ["FPGA"] semiconductor) chip, such as a chip developed through a register transfer level ("RTL") design process.

[0064] The memory **870** is a non-transitory computer readable medium and includes, for example, a program storage area and a data storage area. The program storage area and the data storage area can include combinations of different types of memory, such as read-only memory ("ROM"), random access memory ("RAM") (e.g., dynamic RAM ["DRAM"], synchronous DRAM ["SDRAM"], etc.), electrically erasable programmable read-only memory ("EEPROM"), flash memory, a hard disk, an SD card, or other suitable magnetic, optical, physical, or electronic memory devices. The electronic processor **865** is connected to the memory **870** and executes software instructions that are capable of being stored in a RAM of the memory **870** (e.g., during execution), a ROM of the memory **870** (e.g., on a generally permanent basis), or another non-transitory computer readable medium such as another memory or a disc. Software included in the implementation of the impulse tool **10** can be stored in the memory **870** of the electronic controller **800**. The software includes, for example, firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. The electronic controller **800** is configured to retrieve from memory and execute, among other things, instructions related to the control processes and methods described herein. The electronic controller **800** is also configured to store power tool information on the memory **870** including operational data, information identifying the type of tool, a unique identifier for the particular tool, and other information relevant to operating or maintaining the impulse tool **10**. The tool usage information, such as current levels, motor speed, motor acceleration, motor direction, number of impacts, may be captured or inferred from data output by the sensors **820**. Such power tool information may then be accessed by a user with an external device. In other constructions, the electronic controller **800** includes additional, fewer, or different components.

[0065] The wireless communication controller **850** is coupled to the electronic controller **800** or integrated into the electronic controller **800**. In the illustrated embodiment, the wireless communication controller **850** is located near the foot of the impulse tool **10** to save space and ensure that the magnetic activity of the motor **22** does not affect the wireless communication between the impulse tool **10** and the external device.

[0066] FIG. **9** illustrates a process **900** for controlling the impulse tool **10**. The process **900** begins with the determining a temperature based on one or more signals from the temperature sensors **400**, **405**, **515**, **615**, **700** (STEP **905**). In some embodiments, the determined temperature correlates to a temperature of a fluid within the impulse mechanism (e.g., impulse assembly **18**). If the determined temperature is greater than a temperature threshold at STEP **910**, the controller **800** controls the impulse tool **10** based on the determined temperature (e.g., reduces an output of the impulse tool **10**, shuts down the impulse tool **10**, or otherwise disables operation of the impulse tool **10**) (STEP **915**). In some embodiments, when multiple temperature sensors are used to determine the temperature correlated to the fluid in the impulse mechanism, one temperature sensor can be weighted more heavily than the other. For example, the temperature sensors **405**, **615**, which are outside of the airflow path of the impulse tool **10** can be weighted more heavily (e.g., as a more reliable representation of fluid temperature).

[0067] FIG. **10** illustrates a process **1000** for controlling the impulse tool **10** based on one or more signals from temperature sensors positioned away from a path of airflow circulation in the impulse tool **10**. The process **1000** begins with the determining a temperature based on one or more signals from the temperature sensors **405**, **615** that are at a distal location on the impulse tool **10** away from a path of airflow circulation in the impulse tool **10** (STEP **1005**). In some embodiments, the determined temperature correlates to a temperature of a fluid within the impulse mechanism (e.g., impulse assembly **18**). In some embodiments, the controller **800** correlates the determined temperature to the temperature of the fluid based on the specific heat capacity of the materials of components positioned between the fluid and the temperature sensors **405**, **615**. If the determined temperature is greater than a temperature threshold at STEP **1010**, the controller **800** controls the impulse tool **10** to shut down or otherwise disable operation of the impulse tool **10** at STEP **1015**. In some embodiments, when the determined temperature is greater than the temperature threshold, the temperature of the fluid within the impulse mechanism is, or is likely, above an acceptable temperature for safe operation of the impulse tool **10**.

[0068] FIG. **11** illustrates a graph **1100** of the operation of the impulse tool **10** based on one or more signals from temperature sensors positioned away from a path of airflow circulation in the impulse tool **10**. A temperature threshold **1105** (e.g., 80° C.) is illustrated. In response to the temperature determined based on one or more signals from temperature sensors positioned away from a path of airflow circulation in the impulse tool **10** reaching the temperature threshold **1105**, the controller **800** controls the impulse tool **10** to shut down or otherwise disables operation of the impulse tool **10**. In some embodiments, the graph **1100** corresponds to control of the impulse tool **10** based on one or more temperature signals from the temperature sensors **405**, **615**. It should be noted that when the determined temperature reaches 80° C., the temperature of the fluid in the impulse assembly **18** may be significantly higher than 80° C.

[0069] FIG. **12** is a process **1200** for controlling the impulse tool **10** based on one or more signals from temperature sensors positioned on the Hall effect sensor PCB **505**. The process **1200** begins with determining a temperature based on one or more signals from the temperature sensors **400**, **515** (STEP **1205**) positioned on a Hall effect sensor PCB (for example, the Hall effect sensor PCB **505**). In some embodiments, the determined temperature correlates to a temperature of a fluid within the impulse mechanism (e.g., impulse assembly **18**). However, factors may influence the temperature determined at STEP **1205**, preventing a direct correlation between the determined temperature and the temperature of the fluid within the impulse mechanism. For example, one such factor may be airflow generated by the rotor fan blade.

[0070] If the determined temperature is greater than or equal a first temperature threshold at STEP **1110**, the controller **800** controls the impulse tool **10** to shut down or otherwise disable operation of the impulse tool **10** (STEP **1215**). If the determined temperature is less than a first temperature threshold at STEP **1210**, the controller **800** compares the determined temperature to a second temperature threshold (STEP **1220**). If the determined temperature is less than the first temperature threshold but greater than or equal to the second temperature threshold at STEP **1220**, the amount of time for which the determined temperature is less than the first temperature threshold but greater than or equal to the second temperature threshold is compared to a time threshold (STEP **1225**).

[0071] If the time threshold has been reached at STEP **1225**, the controller **800** controls the impulse tool **10** to shut down or otherwise disable operation of the impulse tool **10** (STEP **1215**). In some embodiments, when the impulse tool **10** shuts down or is otherwise disabled, heat may quickly saturate a stator region as heat transfers from a gearcase mechanism toward a cooling fan of the impulse tool **10**. The determined temperature based on signals from the temperature sensors **400**, **515** positioned on the Hall effect sensor PCB **505** may rise to a temperature that is closer to the temperature of the fluid. Stopping the motor **22** generally causes a rise in the determined temperature based on signals from the temperature sensors **400**, **515** positioned on the Hall effect sensor PCB **505**

[0072] If, at STEP **1220**, the determined temperature is not greater than or equal to the second temperature threshold, the process **1200** returns to STEP **1205**. If, at STEP **1225**, the time threshold has not been satisfied, the process **1200** returns to STEP **1205**. In some embodiments, when the determined temperature is less than the first temperature threshold but greater than or equal to the second temperature threshold and the time threshold has not been satisfied, the controller **800** drives the motor **22**, impulse assembly **18**, or both, harder or faster, respectively, to compensate for a loss of drivetime when the impulse tool **10** is shut down due to the time threshold being satisfied. In some embodiments, controller **800** increases the speed of the motor **22**, drives the impulse assembly **18** harder, or both, to compensate for changes in the performance of the impulse tool **10** due to an increase in the temperature of the fluid in the impulse assembly **18**. In some embodiments, the controller **800** may calculate an optimal torque, an optimal speed, or both, for compensating for performance loss due to the viscosity of the fluid changing as the temperature of the fluid changes. The controller **800** may increase the speed of the motor **22**, drive the impulse assembly **18** harder, or both, by implementing field weakening (e.g., controlling a conduction angle, controlling phase advance, implementing negative id injection in a field-oriented control motor control technique, etc.

[0073] In some embodiments, when the determined temperature is less than the first temperature threshold but greater than or equal to the second temperature threshold and the time threshold has not been satisfied, the controller **800** notifies a user that the tool may be too hot via, for example, one or more of the indicators **1140**. For example, a notification that the impulse tool **10** may be too hot includes the motor **22** coasting to a stop, outputting a noise, blinking a light (for example, a work light LED **1145**), a combination of the foregoing, or the like.

[0074] In some embodiments, the time threshold utilized in the process **1200** is a greater amount of time than the amount of time that applications of the impulse tool **10** typically take. Therefore, a user may turn off or shut down the impulse tool **10** before the time threshold is ever satisfied at STEP **1225**. However, the amount of time the impulse tool **10** needs to complete an application or task may increase when the temperature of the fluid in the impulse assembly **18** is high. In some embodiments, instead of a time threshold, the process **1200** may utilize a threshold number of rotations of the motor **22**, a threshold number of cycles of the impulse tool **10**, or the like.

[0075] FIG. **13** illustrates a graph **1300** of the operation of the impulse tool **10** based on the temperature determined based on one or more signals from temperature sensors positioned on the Hall effect sensor PCB **505**. A first temperature threshold **1305** (e.g., 75° C.) and a second temperature threshold **1310** (e.g., 60° C.) are illustrated. At point A, the controller **800** determines that the determined temperature is greater than the second threshold but less than the first threshold. In such a case, a timer can be started and compared to a time threshold (e.g., 30 seconds). If the determined temperature remains between the two temperature thresholds for the time threshold, the impulse tool **10** can be, for example, shut down at point B. In some embodiments, hysteresis can be used to control thermal shutdown based on a loss of heat. At point C, the determined temperature reaches the first temperature threshold **1305**, and the controller **800** controls the impulse tool **10** to, for example, shut down to force cooling at point D (e.g., a required amount of temperature drop [e.g., 15° C.] before the impulse tool **10** can be restarted). In some embodiments, cooling of the impulse tool **10** may be performed by a rotor fan. In some implementations, active cooling solutions auxiliary to the movement of the motor **22**, such as additional fans, thermo-electric coolers, liquid cooling pumps, or the like, can be employed to cool the impulse tool **10**. In some embodiments, the graph **1300** corresponds to the control of the impulse tool **10** based on one or more temperature signals from the temperature sensors **400**, **515**.

[0076] As illustrated in FIG. **11** and FIG. **13** by the temperature threshold **1105** of 80° C. and the first temperature threshold **1305** of 75° C., respectively, a temperature threshold may be higher when a determined temperature is based on one or more signals from temperature sensors positioned away from a path of airflow circulation in the impulse tool **10** than a temperature threshold is when the determined temperature is based on one or more signals from temperature sensors positioned on, for example, the Hall effect sensor PCB **505**. For example, temperature sensors positioned on the Hall effect sensor PCB **505** are more easily influenced by external factors, such as airflow, than from temperature sensors positioned away from a path of airflow circulation in the impulse tool **10**.

[0077] The processes **1000**, **1200** can be implemented independently of one another or in

conjunction with one another. As a result, one process **1000**, **1200** can function as a failsafe for the other process **1000**, **1200** in the event, for example, of the failure of a temperature sensor. In another example, measurements made by temperature sensors **400**, **515** positioned on a Hall effect sensor PCB **505** may be more easily influenced by external factors than measurements made by temperature sensors **405**, **615** positioned away from a path of airflow circulation in the impulse tool **10**. For example, when a user of the impulse tool **10** uses compressed air to cool the impulse tool **10**, the fluid in the impulse assembly **18** and temperature sensors **405**, **615** positioned away from a path of airflow circulation in the impulse tool **10** may not be cooled by the compressed air as quickly as temperature sensors **400**, **515** positioned on a Hall effect sensor PCB **505**. Therefore, in this example, a temperature determined based on one or more signals from the temperature sensors **405**, **615** positioned away from a path of airflow circulation in the impulse tool **10** may be a more accurate reflection of the temperature of the fluid in the impulse assembly **18** than a temperature determined based on one or more signals from the temperature sensors **400**, **515** positioned on a Hall effect sensor PCB **505**. In some embodiments, the temperature thresholds described herein are static values (e.g., 75° C.). In some embodiments, the temperature sensors **400**, **405**, **515**, **615**, **700** can be used to measure temperature regardless of whether the trigger **1130** is being pulled.
[0078] In some embodiments, the controller **800** is configured to determine a relationship between heat dissipation and time since a most recent application of the impulse tool **10**. In some embodiments, one or more of the temperature thresholds can be dynamically modified (e.g., based on operational mode, ambient temperature, etc.). For example, the impulse tool **10** may include a real time clock, a coin cell battery, or both, and the controller **800** may adjust one or more of the temperature thresholds based on whether the impulse tool **10** is currently running or recently stopped running, battery life, the duty cycle of an application, a time since the most recent application of the impulse tool **10** completed, and a relationship between heat dissipation and time.
[0079] In some embodiments, when a temperature produces inaccurate or unrealistic readings, the controller **800** may ignore the readings from one or all of the temperature sensors **400**, **405**, **515**, **615**, **700** and operate the impulse tool **10** in an unrestricted manner (without performing processes **900**, **1000**, or **1200**). When operating in an unrestricted manner. the impulse tool **10** may experience thermal failure, but, until the thermal failure occurs, the user's workflow is not interrupted.
[0080] Thus, embodiments described herein provide, among other things, a power tool including an impulse assembly. Various features and advantages are set forth in the following claims.

Claims

1. A power tool comprising: a motor; an impulse assembly configured to be driven by the motor; a temperature sensor configured to output a signal related to a temperature of the power tool, the temperature of the power tool being correlated to a temperature of a fluid within the impulse assembly; and a controller connected to the temperature sensor, the controller configured to: determine the temperature of the power tool based the signal from the temperature sensor, and control, in response to the temperature of the power tool being greater than a temperature threshold, the power tool based on the temperature of the power tool.
2. The power tool of claim 1, wherein, to control the power tool based on the temperature of the power tool, the controller is configured to one of shut down the power tool or reduce an output of the power tool.
3. The power tool of claim 1, wherein the temperature sensor is positioned away from a path of airflow circulation within the power tool.
4. The power tool of claim 3, wherein the temperature sensor is positioned on a light assembly printed circuit board (“PCB”).
5. The power tool of claim 1, wherein the temperature sensor is positioned on a Hall effect sensor

printed circuit board (“PCB”).

6. The power tool of claim 1, wherein the controller is further configured to: compare, in response to the temperature of the power tool being less than the temperature threshold and greater than or equal to a second temperature threshold, an amount of time for which the temperature of the power tool is less than first temperature threshold and greater than or equal to the second temperature threshold to a time threshold; and disable, in response to the amount of time reaching the time threshold, the power tool.

7. A power tool comprising: a motor; an impulse assembly configured to be driven by the motor; a first temperature sensor configured to output a first signal related to a temperature of the power tool, the temperature of the power tool being correlated to a temperature of a fluid within the impulse assembly; a second temperature sensor configured to output a second signal related to the temperature of the power tool, the temperature of the power tool being correlated to the temperature of the fluid within the impulse assembly; and a controller connected to the first temperature sensor and the second temperature sensor, the controller configured to: determine the temperature of the power tool based the first signal from the first temperature sensor and the second signal from the second temperature sensor, and control, in response to the temperature of the power tool being greater than a temperature threshold, the power tool based on the temperature of the power tool.

8. The power tool of claim 7, wherein, to control the power tool based on the temperature of the power tool, the controller is configured to shut down the power tool.

9. The power tool of claim 7, wherein the first temperature sensor and the second temperature sensor are positioned away from a path of airflow circulation within the power tool.

10. The power tool of claim 9, wherein the first temperature sensor is positioned on a light assembly printed circuit board (“PCB”).

11. The power tool of claim 10, wherein the second temperature sensor is positioned on a Hall effect sensor PCB.

12. The power tool of claim 7, wherein the controller is further configured to: compare, in response to the temperature of the power tool being less than the temperature threshold and greater than or equal to a second temperature threshold, an amount of time for which the temperature of the power tool is less than first temperature threshold and greater than or equal to the second temperature threshold to a time threshold; and disable, in response to the amount of time reaching the time threshold, the power tool.

13. A method of controlling a power tool, the method comprising: receiving, from a temperature sensor, a signal related to a temperature of the power tool, the temperature of the power tool being correlated to a temperature of a fluid within an impulse assembly; determining the temperature of the power tool based the signal from the temperature sensor, and controlling, in response to the temperature of the power tool being greater than a temperature threshold, the power tool based on the temperature of the power tool.

14. The method of claim 13, wherein controlling the power tool based on the temperature of the power tool includes one or shutting down the power tool or reducing an output of the power tool.

15. The method of claim 13, wherein the temperature sensor is positioned away from a path of airflow circulation within the power tool.

16. The method of claim 15, wherein the temperature sensor is positioned on a light assembly printed circuit board (“PCB”).

17. The method of claim 15, wherein the temperature sensor is positioned on a Hall effect sensor printed circuit board (“PCB”).

18. The method of claim 13, further comprising: comparing, in response to the temperature of the power tool being less than the temperature threshold and greater than or equal to a second temperature threshold, an amount of time for which the temperature of the power tool is less than first temperature threshold and greater than or equal to the second temperature threshold to a time threshold; and disabling, in response to the amount of time reaching the time threshold, the power tool.

19. The method of claim 13, wherein the temperature sensor is positioned within the impulse assembly.

20. The method of claim 13. further comprising: receiving, from a second temperature sensor, a second signal related to the temperature of the power tool, the temperature of the power tool being correlated to the temperature of the fluid within an impulse assembly.